Seth Ricks ECEN 470 PS03: 3.45(a)(b)(c), 3.46(a)(b)(Hint: see c) 3.53(a), 3.55

1/6

(3.45) Find TF below:

a.) Which meet an overshoot of $M_p \leq 5\%$?
b.) " " a rise time of $t_r \leq 0.5$sec?
c.) " " a setting time of $t_s \leq 2.5$sec?

$$G_1(s) = \frac{40}{s^2+4s+40}$$
$$G_2(s) = \frac{40}{(s+1)(s^2+4s+40)}$$
$$G_3(s) = \frac{120}{(s+3)(s^2+4s+40)}$$
$$G_4(s) = \frac{20(s+2)}{(s+1)(s^2+4s+40)}$$
$$G_5(s) = \frac{36040/401\,(s^2+s+401)}{(s^2+4s+40)(s^2+s+401)}$$

$$H(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$G_1(s)$: $4s = 2\zeta\omega_n s \rightarrow 2 = \zeta\omega_n$

$40 = \omega_n^2$

$\omega_n = \sqrt{40} = 2\sqrt{10}$

$2 = \zeta(2\sqrt{10})$

$\zeta = \frac{1}{\sqrt{10}}$

$t_r = \frac{1.8}{\omega_n} = \frac{1.8}{2\sqrt{10}} = 0.285$ ✓

$t_s = \frac{4.6}{\zeta\omega_n} = \frac{4.6}{(\frac{1}{\sqrt{10}})(2\sqrt{10})} = 2.3$ ✓

$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$

$= e^{-\pi/\sqrt{10}/\sqrt{1-(1/\sqrt{10})^2}}$

$= 0.35 \rightarrow 35\%$ X

$M_p \leq 5\%$? NO
$t_r \leq 0.5$ sec? Yes
$t_s \leq 2.5$ sec? Yes

2/6

$G_2(s) = \frac{40}{(s+1)(s^2+4s+40)}$ Poles: $s = -2 \pm j6$, $s = -1$

* within factor of 4 → significant effect *

For $G_2(s)$, since pole is within factor of 4 of the real part of the original poles, overshoot will decrease, rise time will increase, and settling time will increase.

$M_p \downarrow \leq 5\%$? → probably ✓
$t_r \uparrow \leq 0.5$ sec? → probably not ×
$t_s \uparrow \leq 2.5$ sec? → probably not ×

$G_3(s) = \frac{120}{(s+3)(s^2+4s+40)}$

Original: $6 = -2$
new: $s = -3$
$Re\{p\} < 46$: yes
-3 further left than -2 → faster
$M_p \downarrow \leq 5\%$? → probably ✓
$t_r \uparrow \leq 0.5$ sec? → probably not ×
$t_s \uparrow \leq 2.5$ sec? → probably not ×

$G_4(s) = \frac{20(s+2)}{(s+1)(s^2+4s+40)}$

Original: $6 = -2$ closer to origin
new p: $s = -1$
new z: $s = -2$
} → within factor of 4 = yes
$M_p \downarrow$ → ✓
$t_r \uparrow$ → ×
$t_s \uparrow$ → ×

$G_5(s) = \frac{36040/401\,(s^2+s+4011)}{(s^2+4s+40)(s^2+s+901)}$ Original: $6 = -2$
New poles: $s = -0.5 \pm j30$
New zeros: $z = -0.5 \pm j20$

both -0.5, balance each other out.
M_p ×
t_r ✓ t_s ×

After MATLAB:

3/6

	$M_p \leq 5\%$	$t_r \leq 0.5$ sec	$t_s \leq 2.5$ sec
G_1	×	✓	✓
G_2	✓	×	×
G_3	✓	✓	✓
G_4	✓	×	×
G_5	×	✓	✓
	↑ a.)	↑ b.)	↑ c.)